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Recognition of Diabetic Retinopathy using Deep Learning

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***Abstract*—Retinopathy is the disorder of the retina. Diabetic Retinopathy (DR) is the primary cause of vision loss and blindness. The diagnosis of DR is a difficult and time-consuming process that necessitates skilled medical professionals. The proposed approach aims to identify a specific stage linked to diabetic retinopathy using deep learning approach. The basic DR stages can be identified by the proposed methodology, which includes Mild NPDR (Non-Proliferative Diabetic Retinopathy), Moderate NPDR, Severe NPDR, PDR (Proliferative Diabetic Retinopathy) and No DR. A dataset from Kaggle is utilized to train the model using a Convolutional Neural Network (CNN). The work is carried out on a Google Colab Notebook using Python programming. The proposed methodology yields training and validation accuracy of 86.66% and 75.96% respectively.**

Keywords—retinopathy, diabetic retinopathy, artificial intelligence, convolutional neural network, deep learning, recognition

I. Introduction

The ability to perceive and navigate the environment around us makes the eyes an essential component of the human body. They are essential to daily activities including healthcare and communication. Diabetic Retinopathy (DR) is a serious condition that can harm the eyes. It is characterized by damage to the blood vessels of the retina.

Nearly one-third of persons over 40 with diabetes have diabetic retinopathy, which destroys the retina. DR is commonly characterized by unclear or vacant areas in the vision, spots or black threads floating in the vision (sometimes known as "floaters"), and vision loss. Undiagnosed conditions or inadequate treatment will lead to blindness. Thus, maintaining an individual's eye health depends on a prompt and precise diagnosis. How diabetic eyes differ from healthy eyes is depicted in Fig. 1.

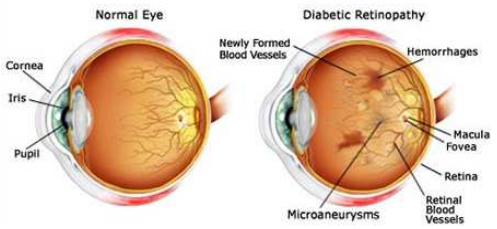

Fig. 1. Healthy and diabetic eye.

Loss of vision is caused by damage to the blood vessels in DR. This results from damage to the retina's blood vessels caused by elevated blood sugar levels. Rarely, new blood vessels might form and cause eyesight loss as well. There are no symptoms in the early stages. The symptoms of DR worsen, including hazy vision, night blindness, faded colors, and vision loss. In particular, adults in their middle and older years are affected by DR. Diabetes' elevated blood glucose levels can damage retinal blood vessels, resulting in blood leakage and vision impairment.

The fundamental stages in diabetic retinopathy are depicted in Fig. 2. The normal eye without DR (Fig. 2A), shows no symptoms or indicators of diabetic retinopathy. In Mild NPDR stage (Fig. 2B), blood or fluid leaks from tiny blood vessels in the retina. This could result in minor retinal edema or the development of tiny deposits known as exudates. At this point, visual impairment or symptoms are frequently absent.

In Moderate NPDR stage (Fig. 2C), areas of the retina receive less blood flow when blood vessels in the retina become clogged. Therefore, the retina could signal the growth of new blood vessels. Severe NPDR stage (Fig. 2D) is characterized by more profound retinal blood channel obstructions. When the retina does not receive enough blood, aberrant blood vessels begin to form and cause problems like retinal hemorrhages or edema. At this point, vision may become considerably impaired.

In PDR (Fig. 2E), the retina produces new blood vessels, however these blood vessels are brittle and prone to bleeding. Further formation of scar tissue may result in retinal detachment and, if untreated, significant vision loss or blindness. As a severe form of diabetic retinopathy, PDR calls for immediate medical attention, frequently in the form of laser treatment or eye injections.

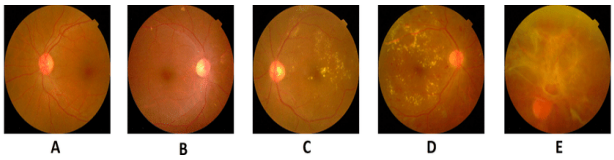

Fig. 2. Fundus images showing five stages of diabetic retinopathy: (A) Without DR, (B) Mild, (C) Moderate, (D) Severe, (E) PDR.

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II. Related Work

Numerous studies on the identification of diabetic retinopathy have been published in the literature, and a number of problem solvers have expressed interest in identifying the various stages of DR. An early diagnosis of DR is essential to minimize its complications. This is because there are so many uses for this rigorous and demanding area in the modern world.

Using the EfficientNet B7 model as a basis, a deep learning method was created to automatically identify diabetic retinopathy using Aptos2019 dataset [1][2]. A refined ResNet50 deep learning model was utilized in a transfer learning strategy to classify diabetic retinopathy, where the results of the experiment demonstrate the efficiency of transfer learning by showing that supervised pre-training on ImageNet, followed by fine-tuning on labelled fundus images, considerably increases the efficacy of the medical image classifier when trained on full training data [3].

A novel method for the automated diagnosis of diabetic retinopathy using deep learning models was employed; it involved preprocessing and preparing retinal images, training UNet for retinal image segmentation, multi-classifying segmented images using a custom-designed CNN, and merging CNN models for improved performance. The method involved transfer learning with Inception network for retinopathy classification [4]. Research on Deep DR, a method that uses deep learning to diagnose all stages of diabetic retinopathy, was conducted to improve the diagnosis process. It provides a graphical tip that aids users in identifying the location and frequency of various abrasion kinds [5].

The stages of diabetic retinopathy were successfully predicted using an enhanced CNN and data augmentation with an accuracy rate of 82% [6][7]. The use of ConvNeXt for an early diabetic retinopathy diagnosis was suggested using the cutting-edge deep learning model ConvNeXt in conjunction with the pre-processing method Contrast Limited Adaptive Histogram Equalization (CLAHE) to provide encouraging results [8]. Using a pre-trained convolutional graph neural network (CGNN) and a set of retinal images from the eyepacs diabetic retinopathy database, it was suggested to automatically identify and classify the symptoms of diabetic retinopathy [9]. A categorization approach was used utilizing U-Net and image processing for the scientific representation of different stages of DR, and the modified Network algorithm's performance was assessed using the DIARETDB0 and Kaggle samples [10] [11].

III. Proposed Methodology

Artificial Intelligence (AI) is one of the hottest technologies in the world nowadays. Using the Kaggle dataset and a deep learning approach, the proposed methodology aims to create an accurate detection system for diabetic retinopathy. Fig. 3 shows the block diagram of the proposed methodology, which is primarily divided into five stages: Phases A through E include creating the dataset, preprocessing the data, splitting the data, designing the CNN, and training and validating the results.

Fig. 3. Block diagram of the proposed approach.

A. Dataset Creation

The dataset used in this work is taken from Kaggle, which mostly includes five stages of diabetic retinopathy. They are: Mild NPDR, Moderate NPDR, Severe NPDR, PDR and No DR. The gathered photos have diverse color spaces (grayscale and color) and varied in size. Pre-processing the dataset is therefore necessary before training the model.

B. Data Pre-processing

During the data pre-processing stage, three main operations are performed: Image Resizing, Image Labeling, and Image Gray-Scaling.

- In Image Resizing stage, using the resize() function of the OpenCV library, all of the dataset's images are resized to 48 (width) x 48 (height).
- In Image Labeling stage, all the images from the dataset are properly labeled with their DR stages. For instance, if the images are of Mild NPDR type, then those are labeled as mild_1, mild_2, mild_3 and so forth.
- In Image Gray-Scaling stage, all of the dataset's images are converted to grayscale using the OpenCV library's cvtColor() method.

The dataset has 3662 images in total, of which the sample images are shown in Fig. 4. The number in parenthesis represents the total number of images for that specific DR stage.

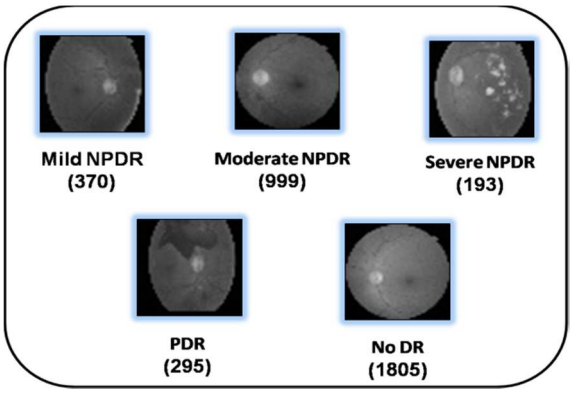

Fig. 4. Sample images from the dataset.

C. Data Splitting

The final dataset for this study is split into 70:30 ratios, which indicates that 30% of the dataset is used for validation and 70% of the dataset is utilized for training. Following data

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splitting, there are 2564 images in the training dataset and 1098 images in the validation dataset. (Fig. 5)

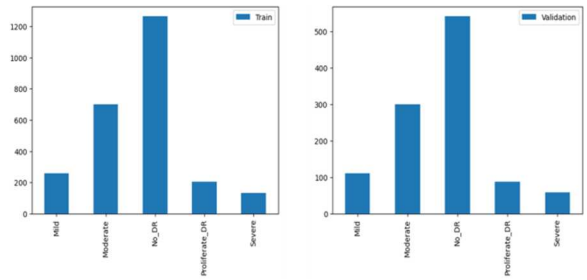

Fig. 5. Splitting of training and validation dataset.

D. CNN Design

The proposed CNN structure is depicted in Fig. 6, and it is created using the sequential approach. Four convolutional layers with a kernel size of three by three, two max-pooling layers with a pool size of two by two, one flattening layer, and two dense layers make up the CNN structure. Lastly, the DR stage is classified using the Softmax activation function.

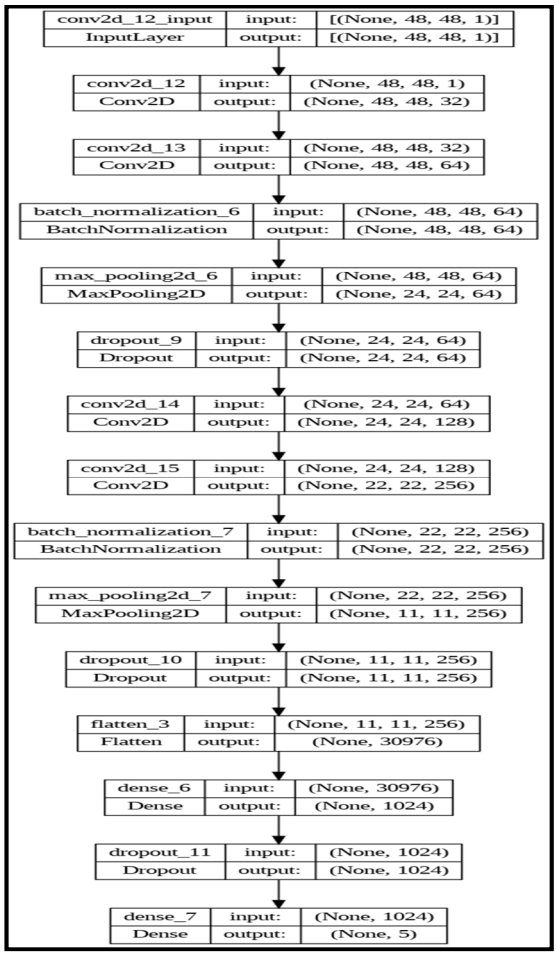

Fig. 6. Proposed architecture of CNN.

Fig. 7 shows the summary of the proposed CNN architecture.

```
Model: "sequential_3"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 conv2d_12 (Conv2D)          (None, 48, 48, 32)        320

 conv2d_13 (Conv2D)          (None, 48, 48, 64)        18496

 batch_normalization_6 (Bat  (None, 48, 48, 64)        256
 chNormalization)

 max_pooling2d_6 (MaxPoolin  (None, 24, 24, 64)        0
 g2D)

 dropout_9 (Dropout)         (None, 24, 24, 64)        0

 conv2d_14 (Conv2D)          (None, 24, 24, 128)       73856

 conv2d_15 (Conv2D)          (None, 22, 22, 256)       295168

 batch_normalization_7 (Bat  (None, 22, 22, 256)       1024
 chNormalization)

 max_pooling2d_7 (MaxPoolin  (None, 11, 11, 256)       0
 g2D)

 dropout_10 (Dropout)        (None, 11, 11, 256)       0

 flatten_3 (Flatten)         (None, 30976)             0

 dense_6 (Dense)             (None, 1024)              31720448

 dropout_11 (Dropout)        (None, 1024)              0

 dense_7 (Dense)             (None, 5)                 5125

=================================================================
Total params: 32114693 (122.51 MB)
Trainable params: 32114053 (122.51 MB)
Non-trainable params: 640 (2.50 KB)
_________________________________________________________________
```

Fig. 7. Summary of the proposed CNN architecture.

E. Training and Validation Phase

The parameters that are specified during CNN's training phase are highlighted in Table I. In this phase, the prediction model is developed using the training data, and its performance is assessed using the validation data.

CNN uses callback procedures like CSVLogger(), ReduceLROnPlateau(), and ModelCheckpoint() during training. After the training phase completion, a prediction model (.h5 file) is produced and utilized to forecast various DR stages.

TABLE I. TRAINING PARAMETERS OF CNN ARCHITECTURE

Sl. No.	Parameter	Value
1.	Batch Size	8
2.	Number of Epochs	60
3.	Learning Rate	0.0001
4.	Metric	Accuracy
5.	Optimizer	Adam

The accuracy and loss rate for the first ten epochs during the training and validation phase are shown in Fig. 8.

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```
Epoch 1/60
320/320 [==============================] - ETA: 0s - loss: 5.0472 - accuracy: 0.5286
Epoch 1: val_loss improved from inf to 4.37151, saving model to Model.h5
320/320 [==============================] - 17s 48ms/step - loss: 5.0472 - accuracy: 0.5286 - val_loss: 4.3715 - val_accuracy: 0.4945 - lr: 1.0000e-04
Epoch 2/60
319/320 [==============================>.] - ETA: 0s - loss: 3.5694 - accuracy: 0.6075
Epoch 2: val_loss improved from 4.37151 to 3.28063, saving model to Model.h5
320/320 [==============================] - 18s 55ms/step - loss: 3.5677 - accuracy: 0.6080 - val_loss: 3.2806 - val_accuracy: 0.6953 - lr: 1.0000e-04
Epoch 3/60
319/320 [==============================>.] - ETA: 0s - loss: 3.3522 - accuracy: 0.6311
Epoch 3: val_loss improved from 3.28063 to 3.11114, saving model to Model.h5
320/320 [==============================] - 16s 49ms/step - loss: 3.3509 - accuracy: 0.6318 - val_loss: 3.1111 - val_accuracy: 0.7035 - lr: 1.0000e-04
Epoch 4/60
319/320 [==============================>.] - ETA: 0s - loss: 3.1481 - accuracy: 0.6593
Epoch 4: val_loss improved from 3.11114 to 3.01582, saving model to Model.h5
320/320 [==============================] - 17s 54ms/step - loss: 3.1470 - accuracy: 0.6596 - val_loss: 3.0158 - val_accuracy: 0.7035 - lr: 1.0000e-04
Epoch 5/60
320/320 [==============================] - ETA: 0s - loss: 2.9648 - accuracy: 0.6749
Epoch 5: val_loss improved from 3.01582 to 2.84041, saving model to Model.h5
320/320 [==============================] - 17s 50ms/step - loss: 2.9648 - accuracy: 0.6749 - val_loss: 2.8404 - val_accuracy: 0.7089 - lr: 1.0000e-04
Epoch 6/60
319/320 [==============================>.] - ETA: 0s - loss: 2.7948 - accuracy: 0.6762
Epoch 6: val_loss improved from 2.84041 to 2.67092, saving model to Model.h5
320/320 [==============================] - 17s 54ms/step - loss: 2.7960 - accuracy: 0.6764 - val_loss: 2.6709 - val_accuracy: 0.6998 - lr: 1.0000e-04
Epoch 7/60
319/320 [==============================>.] - ETA: 0s - loss: 2.6093 - accuracy: 0.6817
Epoch 7: val_loss improved from 2.67092 to 2.49762, saving model to Model.h5
320/320 [==============================] - 17s 52ms/step - loss: 2.6110 - accuracy: 0.6819 - val_loss: 2.4976 - val_accuracy: 0.6989 - lr: 1.0000e-04
Epoch 8/60
320/320 [==============================] - ETA: 0s - loss: 2.4275 - accuracy: 0.6827
Epoch 8: val_loss did not improve from 2.49762
320/320 [==============================] - 13s 42ms/step - loss: 2.4275 - accuracy: 0.6827 - val_loss: 2.5188 - val_accuracy: 0.5903 - lr: 1.0000e-04
Epoch 9/60
320/320 [==============================] - ETA: 0s - loss: 2.2706 - accuracy: 0.6882
Epoch 9: val_loss improved from 2.49762 to 2.22015, saving model to Model.h5
320/320 [==============================] - 15s 46ms/step - loss: 2.2706 - accuracy: 0.6882 - val_loss: 2.2202 - val_accuracy: 0.6943 - lr: 1.0000e-04
Epoch 10/60
319/320 [==============================>.] - ETA: 0s - loss: 2.1210 - accuracy: 0.6979
Epoch 10: val_loss improved from 2.22015 to 2.05868, saving model to Model.h5
320/320 [==============================] - 15s 48ms/step - loss: 2.1228 - accuracy: 0.6969 - val_loss: 2.0587 - val_accuracy: 0.6943 - lr: 1.0000e-04
```

Fig. 8. A snapshot of the first ten Epochs.

IV. Results and Observations

In this work, the Python programming language is utilized to train and assess the model's performance using a Google Colab notebook. CNN training makes use of the Tensorflow and Keras libraries. The accuracy measure is used to assess the prediction model's performance.

The prediction model's accuracy throughout training and validation across all 60 epochs is shown in Fig. 9. According to the experiment's findings, the accuracy of training and validation is 86.66% and 75.96%, respectively.

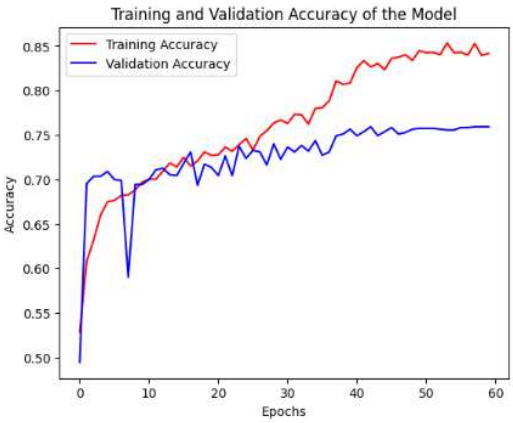

Fig. 9. Training and validation accuracy of the model.

The prediction model's training and validation losses over the course of 60 epochs are shown in Fig. 10. According to the experimental findings, the training and validation losses are 0.4372 and 0.8427 respectively.

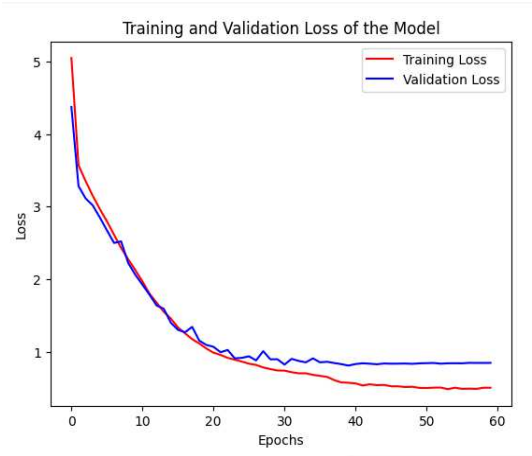

Fig. 10. Training and validation loss of the model.

The above experimental results are tabulated in the Table II.

TABLE II. Experimental Results

Sl. No.	Performance Metrics	Training	Validation
1.	Accuracy	86.66%	75.96%
2.	Loss	0.4372	0.8427

The validation loss measure evaluates a deep learning model's performance on the validation set, where as the training loss measure evaluates how well a deep learning model matches the training data.

The accuracy is determined as, the number of accurate forecasts divided by the total number of predictions the model made. The equation (1) is used to determine the model's accuracy through a confusion matrix.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{1}$$

The Confusion Matrix and Classification Report on validation set are given in Fig. 11:

```
Confusion Matrix:
[[ 18  40  47   3   3]
 [ 37  86 157   8  12]
 [ 44 178 291  14  14]
 [  9  25  51   1   2]
 [  1  23  29   1   4]]


Classification Report:
                precision    recall  f1-score   support

          Mild       0.17      0.16      0.16       111
      Moderate       0.24      0.29      0.26       300
         No_DR       0.51      0.54      0.52       541
Proliferate_DR       0.04      0.01      0.02        88
        Severe       0.11      0.07      0.09        58
```

Fig. 11. Confusion Matrix and Classification Report on validation set.

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V. Conclusion

Using CNN architecture, the proposed approach for identifying diabetic retinopathy produces excellent results. The experimental results highlight the effectiveness of the proposed methodology when compared to previous studies; training and validation accuracy rates are estimated as 86.66% and 75.96% respectively. As a result, the proposed methodology may aid medical professionals in more precisely identifying the DR stage for an early diagnosis. In the future, this method can be integrated into any cloud-based system and linked to an embedded device to gather data in real time.

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